Scratch: Make a Catch Game, Step by Step

Build your first complete game: move, catch, score, lives and game over.

Scratch | Beginner to Intermediate | Ages 7–13

The Big Idea

- How do I make a game in Scratch? Start with a very small game and add one feature at a time.
- A catch game has a player (a basket), a falling object (an apple), a score and a way to lose.
- Plan first: What does the player do? What gives points? What ends the game?
- Test after every step. Fix one thing at a time.

What You Need

- Sprite 1: Basket (the player). Sprite 2: Apple (the falling object).
- Backdrop: any outdoor scene.
- Variables: `score` and `lives`, both For all sprites.
- Sounds: Pop for catching, Boing for missing. Optional.

Build Plan

- Step 1: Move the basket with the mouse or arrow keys.
- Step 2: Make the apple fall from a random place at the top.
- Step 3: Catch it to earn a point and restart it at the top.
- Step 4: Missing it costs a life.
- Step 5: When lives reach 0, say Game over and stop.
- Step 6: Polish with sounds, faster speed and more apples.

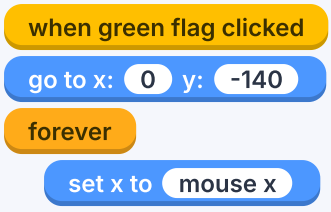
Troubleshooting

- Apple never reaches the basket: check the basket's y position is low enough, near -140.
- Score goes up too fast: add `go to` the top right after catching, so it stops touching.
- Game never ends: make sure the lives check is inside the forever loop.
- Basket jumps: use `set x to (mouse x)`, not `go to` with a y that changes.

💻 The Scripts

Step 1: Basket follows the mouse

Scratch blocks

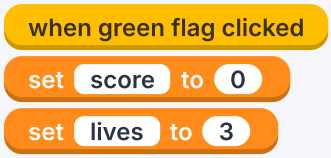


▶ What happens:

The basket slides left and right with the mouse and stays at the bottom.

Step 2: Variables at the start (any sprite)

Scratch blocks

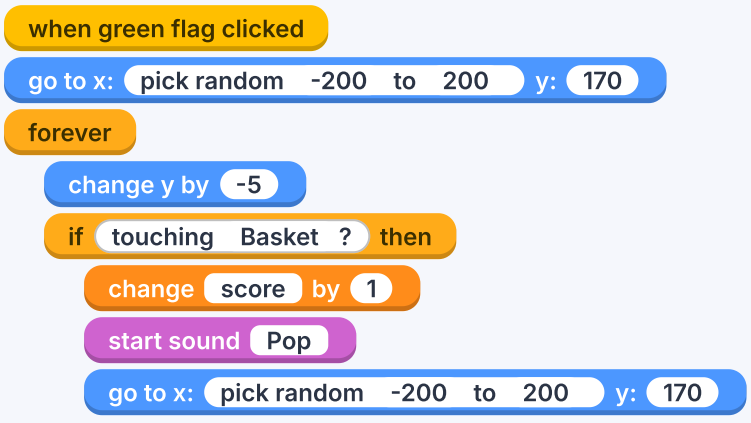


▶ What happens:

Every game starts with 0 points and 3 lives.

Step 3: Apple falls

Scratch blocks

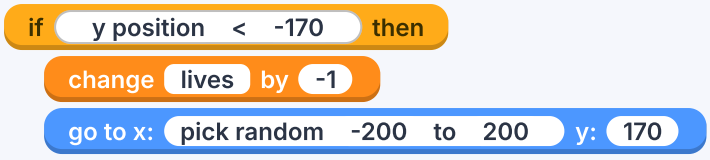


▶ What happens:

The apple falls. Catching it adds 1 point, plays a pop and sends it back to the top.

Step 4: Missing costs a life

Scratch blocks

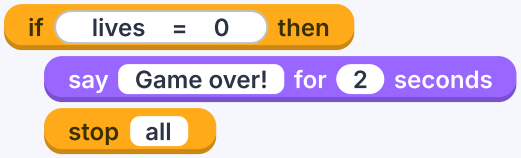


▶ What happens:

Add this inside the same forever loop. An apple that reaches the floor takes a life.

Step 5: Game over

Scratch blocks

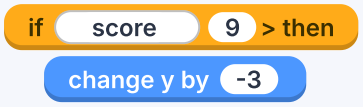


▶ What happens:

Add this inside the forever loop too. At 0 lives the game ends.

Step 6: Get harder

Scratch blocks



▶ What happens:

Add inside the loop. After 10 points the apple falls faster.

📚 Key Terms

Game loop
The forever loop that keeps the game running.

Score
A variable that counts points.

Lives
A variable that counts chances.

Collision
When two sprites touch.

Random
A number chosen by chance.

Difficulty
How hard the game is.

Playtest
Playing your own game to find problems.

Game over
When the game ends.

Prototype
A small, early version of a game.

Polish
Extra touches like sound and effects.

🧮 Quick Reference

```
set x to (mouse x)
```

```
change y by (-5)
```

```
if <touching [Basket]?> then
```

```
change [score] by (1)
```

```
change [lives] by (-1)
```

```
go to x: (pick random (-200) to (200))
y: (170)
```

```
if <(lives) = (0)> then
```

```
stop [all]
```

✏️ Worked Examples

Q: Make the apple fall faster over time.

A: Add a variable called speed. Use change y by ((0) - (speed)) and increase speed by 1 every time the score goes up by 5.

Q: Add a second falling object that costs a life when caught.

A: Duplicate the apple sprite, rename it Bomb. When it touches the basket, change [lives] by (-1) instead of adding to the score.

Q: Add a high score.

A: Make a variable called high score. At game over, if score is bigger than high score, set high score to score.

🚫 Common Mistakes

- Building everything at once, then not knowing which part is broken.
- Forgetting to reset `score` and `lives` when the green flag is clicked.
- Putting the lives check outside the loop, so it only runs once.
- Not moving the apple away after a catch, so it scores again and again.
- Making the game too hard for a first version.

👨‍👩‍👧 Parent Corner

- Play the finished game together and ask what to change to make it harder.
- Ask your child to explain the game as three rules: how you win points, how you lose, and when it ends.
- Encourage a test after every step instead of at the end.
- Ask for a new sprite and a new rule, like a golden apple worth 5 points.
- Share it on the Scratch site together only if the account is set up by a parent.

🎯 Quick Check

1. What are the 2 variables in a catch game?

✅ score and lives.

2. Which block makes the basket follow the mouse?

✅ set x to (mouse x) inside a forever loop.

3. How does the apple restart?

✅ go to a random x at y 170.

4. What ends the game?

✅ When lives reach 0, using stop all.

5. Why test after every step?

✅ So you find and fix one problem at a time.